■ **Tentative Specification**
☐ **Preliminary Specification**
☐ **Approval Specification**

Customer：_____

Product Model: _____

Version : A0

CUSTOMER APPROVED		
Designed by	**Checked by**	**Approved by**

APPROVED		
Prepared by	**Checked by**	**Approved by**

Revision History

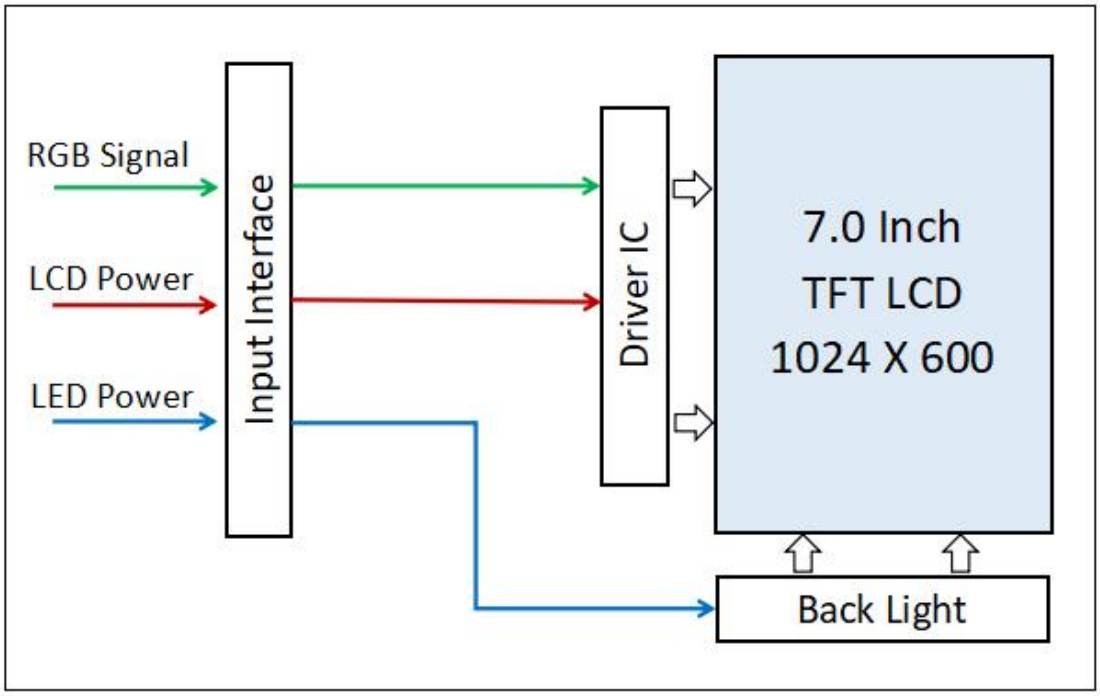
Version	Contents	Date	Note
A0	Original	20240801	-
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1.General Description

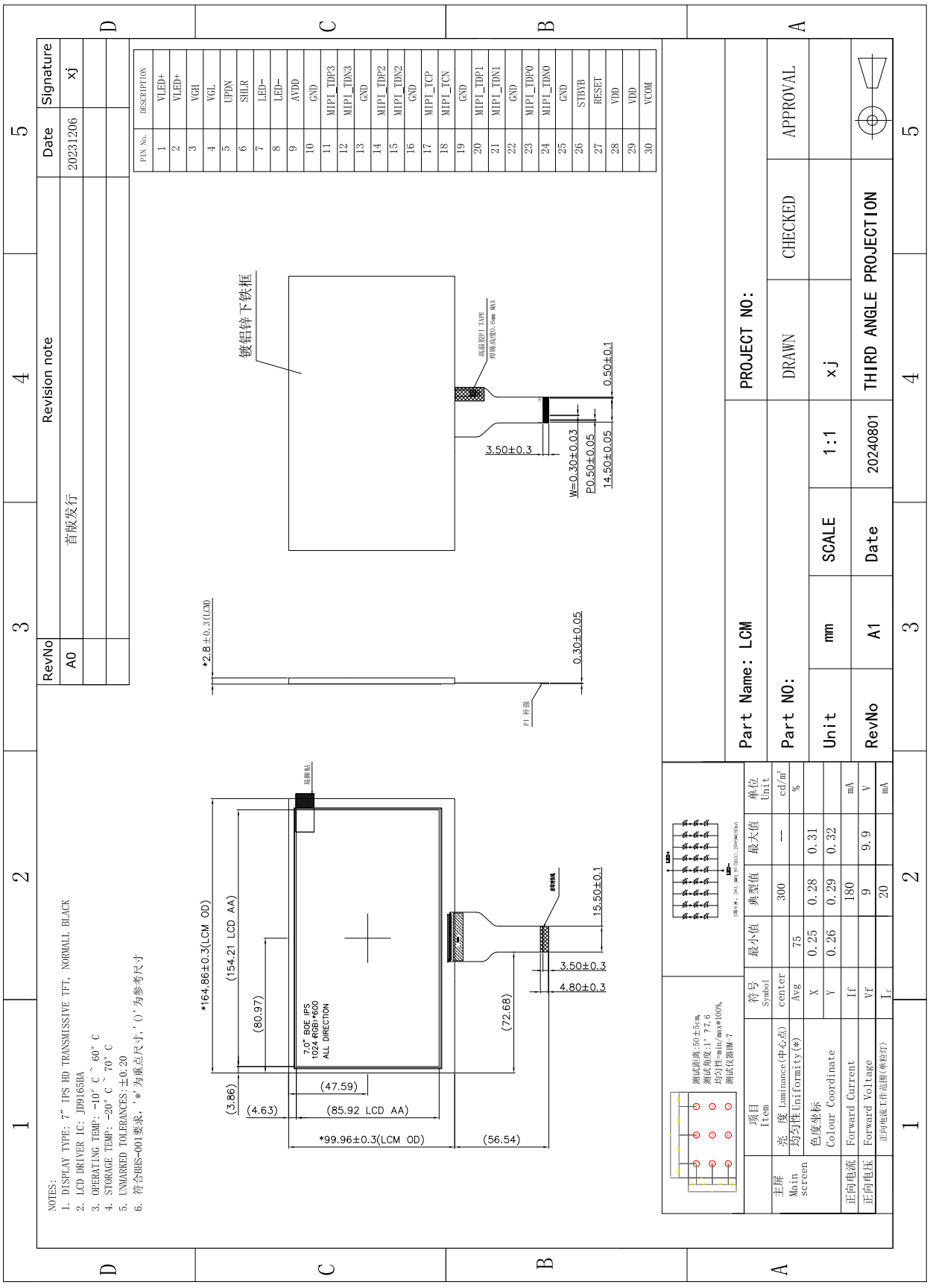
The **LCM-070WS209-A0** is a Color Active Matrix Liquid Crystal Display adopts one integral backlight with High brightness 27-lamps white LED. This LCD has 7 inches diagonally measured active display area with 1024*600 resolution. With a 8bit bit gray scale signal for each dot and 16.7M colors,With BOE Cell.



2. Technology Features

No.	Item	Specification
1	LCD size	7.0 inch
2	Resolution	1024(RGB)X600
3	Display mode	Normally Black
4	Pixel pitch	0.0502(H) X RGB X 0.1432(V) mm
5	Active area	154.21(H)X85.92 (V) mm
6	Module size	163.86 (H)X99.96(V)X2.8(T)mm
7	Pixel arrangement	RGB-stripe
8	Surface Treatment	AG
9	Interface	MIPI
10	Viewing Direction	Full Viewing
11	NTSC	50%
12	Backlight	White LED
13	LCD Driver IC	JD9165BA

3.Outline Dimension



4.Interface Definition

Pin No.	Symbol	Description	Remark
1	LEDA	LED anode	
2	LEDA	LED anode	
3	VGH	Gate ON Voltage	
4	VGL	Gate OFF Voltage	
5	UPDN	Vertical scan direction	
6	SHLR	Horizontal scan direction	
7	LEDK	LED CATHODE	
8	LEDK	LED CATHODE	
9	AVDD	Power for Analog Circuit	
10	GND	Ground	
11	MIPI_TDP3	MIPI differential data input	
12	MIPI_TDN3	MIPI differential data input	
13	GND	GND	
14	MIPI_TDP2	MIPI differential data input	
15	MIPI_TDN2	MIPI differential data input	
16	GND	GND	
17	MIPI_TCP	MIPI differential clock input	
18	MIPI_TCN	MIPI differential clock input	
19	GND	GND	
20	MIPI_TDP1	MIPI differential data input	
21	MIPI_TDN1	MIPI differential data input	
22	GND	GND	
23	MIPI_TDP0	MIPI differential data input	
24	MIPI_TDN0	MIPI differential data input	
25	GND	GND	
26	STBYB	Standby Mode select pin.	
27	RESET	Global reset pin	
28	VDD	Power for Digital Circuit	
29	VDD	Power for Digital Circuit	
30	VCOM	Common-mode voltage	

5. Operation Specifications

5.1 Absolute Max. Rating

The following are maximum values which，if exceeded，may cause faulty operation or damage to the unit.

Item	Symbol	Min.	TYP	Max.	Unit
Supply power voltage	VDD	-0.3	–	3.96	V
I/O power voltage	VDDIO	-0.3	-	3.96	V
AVDD input power	AVDD	-0.3	-	12	V
Gate On Power Supply Voltage	VGH	-0.3	-	VGL+32	V
Gate Off Power Supply Voltage	VGL	VGH-32	-	0.3	V
VMID voltage	VMID	-0.3	-	6.6	V
Common Power Supply Voltage	VCOM_OP	-0.3	-	5.43	V
VOTP (OTP power)	VOTP	-	-	9	V
Operating Temperature	TOPR	-10	-	60	℃
Storage temperature	TSTG	-20	-	70	℃

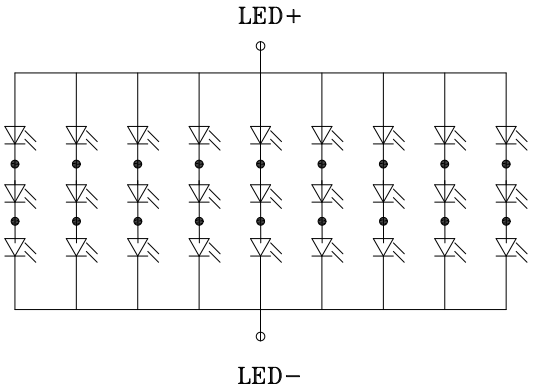
5.2 Typical Operation Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply power voltage	VDD	1.7	1.8	2.0	V
I/O power voltage	VDDIO	1.7	1.8	2.0	V
AVDD input power	AVDD		9.8		V
Gate On Power Supply Voltage	VGH	18		20	V
Gate Off Power Supply Voltage	VGL	-8.5		-7.5	V
Common Power Supply Voltage	VCOM		3.3		V

1. The measuring position is the connector of LCM and the test conditions are under 25℃, Frame Rate=60Hz
2. Vcom must be adjusted to optimize display quality, as Crosstalk and Contrast Ratio etc.. The value is just the reference value. The customer can optimize the setting value by the different D-IC.

5.3 LED Back Light Specification

Item	Symbol	Values			Unit	Remark
		Min.	Typ.	Max.		
LED Current	ILED		180		mA	Total LED
Forward Voltage	VF	8.1	9.0	9.9	V	IF=180mA
LED life time		20000			Hr	

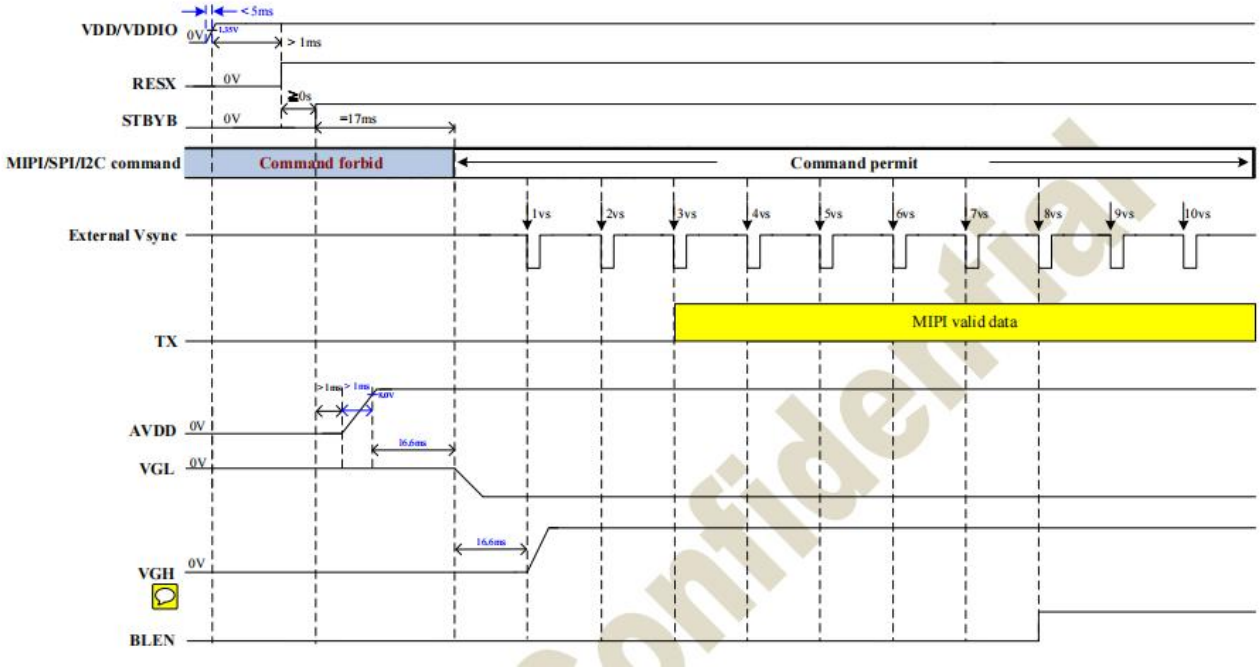


3串9并， 3*3.3=9.9V(MAX),20*9=180mA

5.4 Power On/Off Sequence

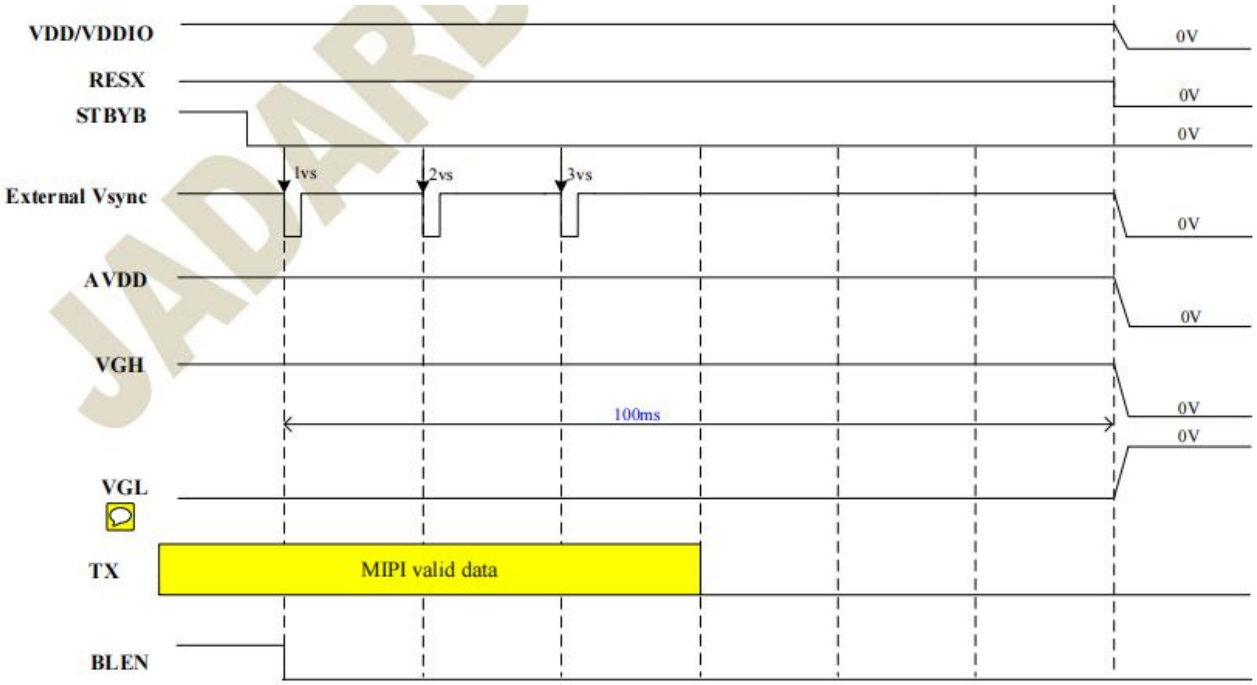
To prevent the device damage from latch up, the power on/off sequence shown below must be followed.

a. Power on



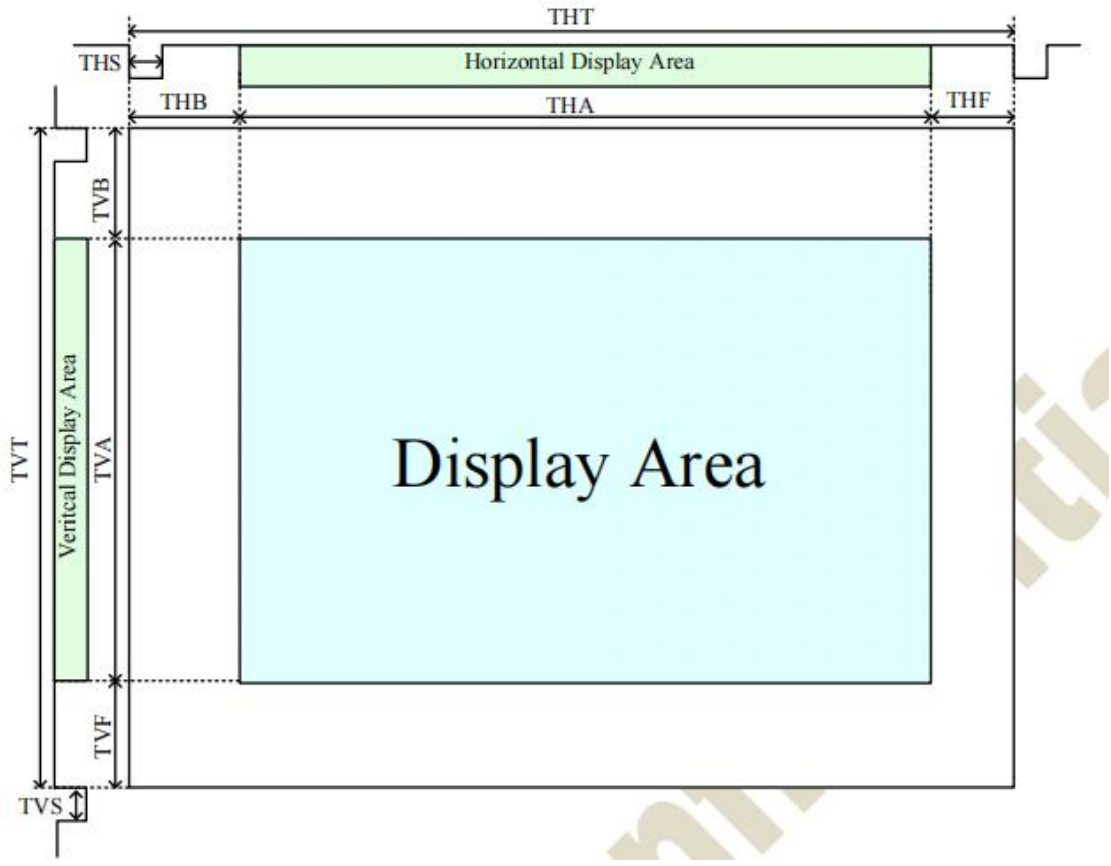
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b. Power off



6. MIPI video input Timing

6.1 MIPI video input timing



6.2 Timing Setting Table

Parameter		Symbol	Min.	Typ.	Max.	Unit
Frame Rate		/	-	60	-	Hz
Clock frequency		1/Tclock	100	-	500	Mbps
Horizontal Section	Period	Tht	1114	1344	1400	Tclock
	Active	THDISP	1024			
	Blanking	THFP	30	160	216	
		THBP	60	160	160	
		THS	1	24	HBP-1	
Vertical Section	Period	Tvt	610	635	800	Tline
	Active	TVDISP	600			
	Blanking	TVFP	2	12	100	
		TVBP	8	23	100	
		TVS	1	2	VBP-1	

7.Optical specifications

Optical characteristics are determined after the unit has been 'ON' and stable for approximately 5 minutes in a dark environment at 25℃.

Item		Symbol	Min.	Typ.	Max.	Unit	Remark
Contrast Ratio		CR	800	1000	-		Note 1 Note 4
Luminance(Center)		L		300		cd/m2	Note 1 Note 6 Note 7
Luminance Uniformity(9P)		LU	75		-	%	
Color Gamuu			-	50	-	%	
Response Time		TRT	-	30	40	ms	Note 3
Color Chromaticity	White	x	Typ-0.03	0.27	Typ+0.03		Note5
		y		0.275			
	Red	x					
		y					
	Green	x					
		y					
	Blue	x					
		y					
Viewing Angle range	Horizontal	ΘR	-	85	-	Deg.	Note 2
		ΘL	-	85	-	Deg.	
	Vertical	ΘT	-	85	-	Deg.	
		ΘB	-	85	-	Deg.	

Note 1: Definition of viewing angle range

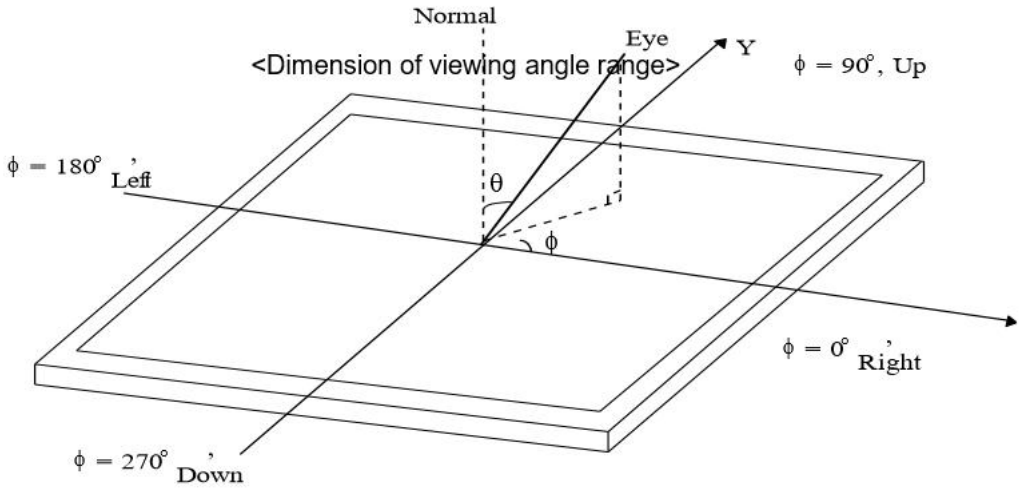


Fig. 4-1 Definition of viewing angle

Note 2: Definition of optical measurement system.

The optical characteristics should be measured in dark room. After 5 minutes operation, the optical properties are measured at the center point of the LCD screen. (Viewing angle is measured by ELDIM-EZ contrast/Height :1.2mm ,Response time is measured by Photo detector TOPCON BM-5A, other items are measured by BM-7A/Field of view: 1° /Height: 500mm.)

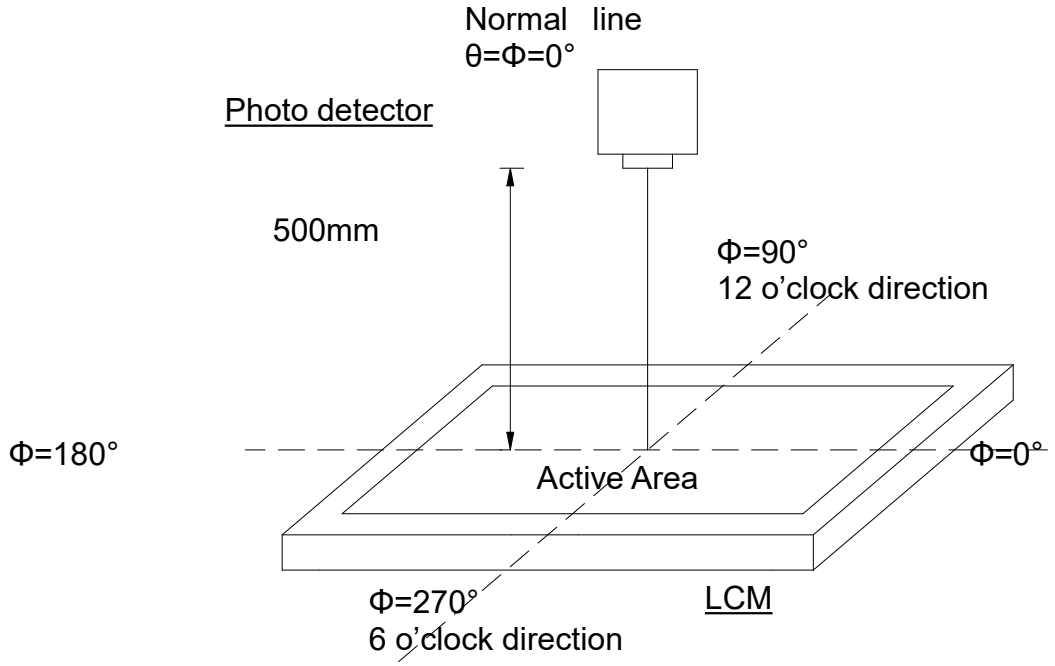


Fig. 4-2 Optical measurement system setup

Note 3: Definition of Response time

The response time is defined as the LCD optical switching time interval between "White" state and "Black" state. Rise time (T_{ON}) is the time between photo detector output intensity changed from 90% to 10%. And fall time (T_{OFF}) is the time between photo detector output intensity changed from 10% to 90%.

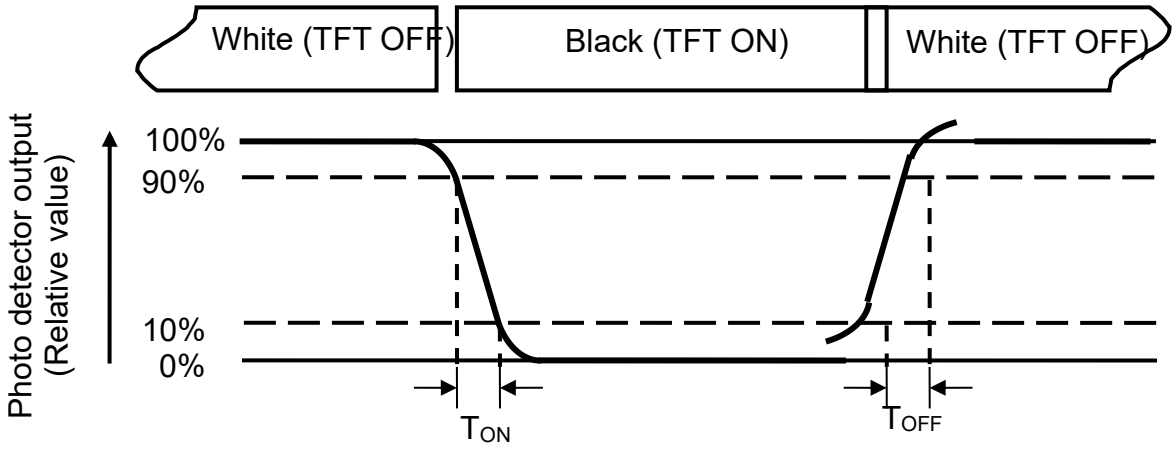


Fig. 4-3 Definition of response time

Note 4: Definition of contrast ratio

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD on the "White" state}}{\text{Luminance measured when LCD on the "Black" state}}$$

Note 5: Definition of color chromaticity (CIE1931)
Color coordinates measured at center point of LCD.
Note 6: All input terminals LCD panel must be ground while measuring the center area of the panel.
Note 7: Definition of Luminance Uniformity
Active area is divided into 9 measuring areas (Refer to Fig. 4-4).Every measuring point is placed at the center of each measuring area.

$$\text{Luminance Uniformity (Yu)} = \frac{B_{min}}{B_{max}}$$

L-------Active area length W----- Active area width

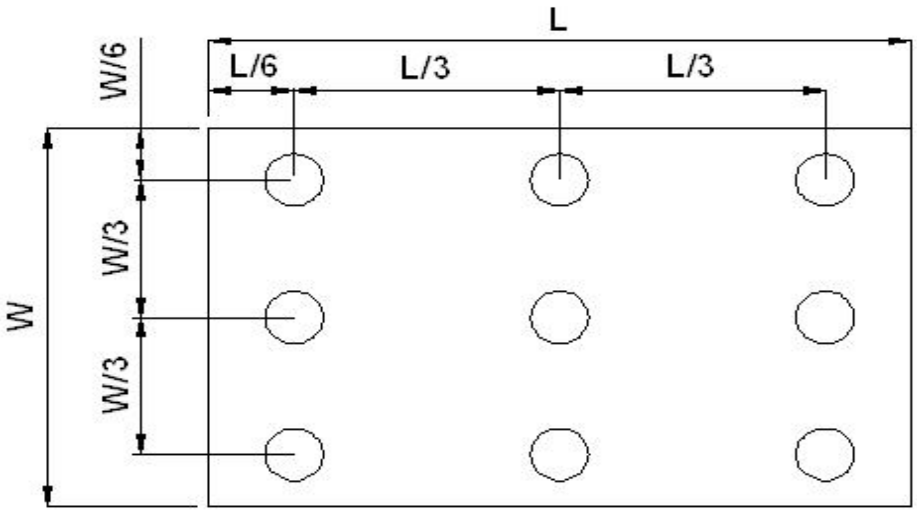


Fig. 4-4 Definition of measuring points

B_{max}: The measured maximum luminance of all measurement position.
B_{min}: The measured minimum luminance of all measurement position.

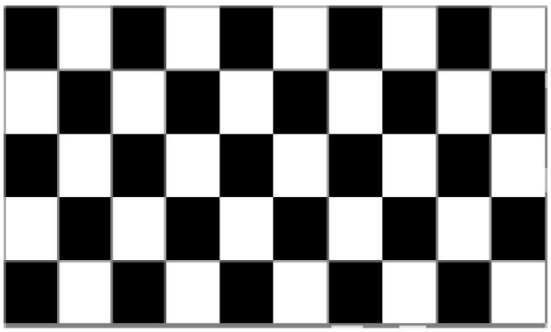
8. Reliability Test Conditions And Methods

Item	Test Conditions	Remark
High Temperature Storage	Ta = 70℃ 48 hrs	
Low Temperature Storage	Ta = -20℃ 48 hrs	
High Temperature Operation	Ts = 60℃ 48 hrs	
Low Temperature Operation	Ta = -10℃ 48hrs	
Operate at High Temperature and Humidity	60℃, 85%RH max. 48 hrs	Operation
Thermal Shock	-10 ℃ /30min~+60 ℃ /30min for a total 24 cycles	Start with cold temperature and end with high temperature
Electrostatic Discharge	Contact=±4KV, class B Air=±8KV, class B **Human Body Mode，150pF/330**	
Image Sticking	**25℃ ; 1hrs**	**Note3**

Note1: In the standard condition, there shall be no practical problem that may affect the display function. After the reliability test, the product only guarantees operation, but don't guarantee all of the cosmetic specification.

Note2: Before cosmetic and function test, the product must have enough recovery time, at least 2 hours at room temperature.

Note3: Condition of image sticking test :25 ℃ ± 2 ℃ , Operation with test pattern sustained for 30min,then change to gray pattern immediately. after 5 mins, the mura must be disappeared completely.

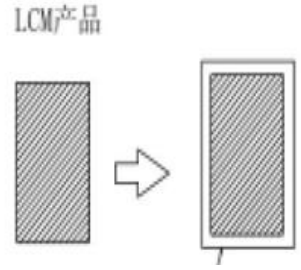


(a) Test Pattern (chess board Pattern)

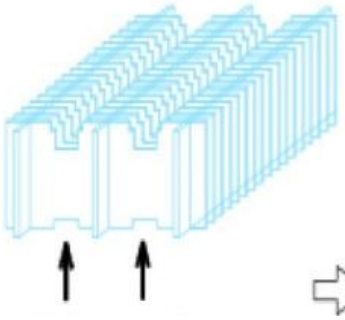


(b) Gray Pattern (127

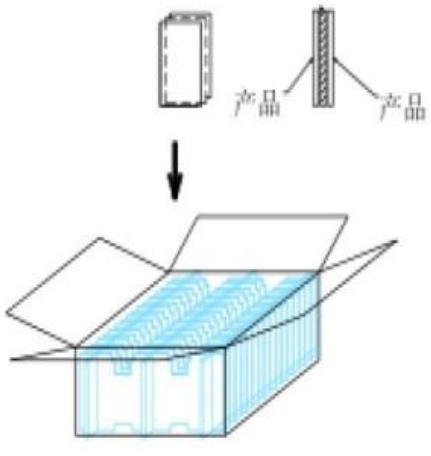
9. Package Drawing



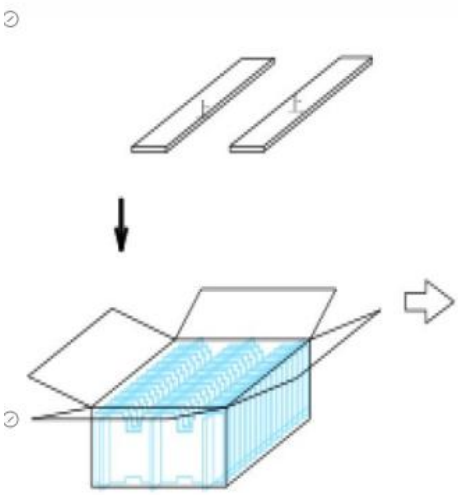
第一步：产品装入袋



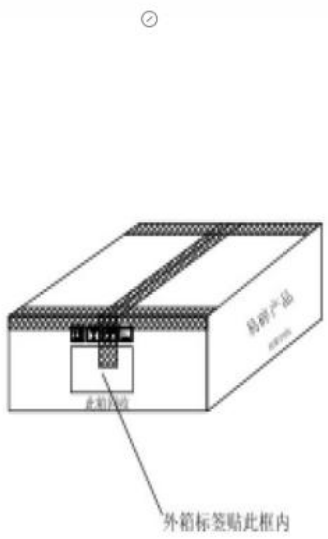
第二步：组装刀卡



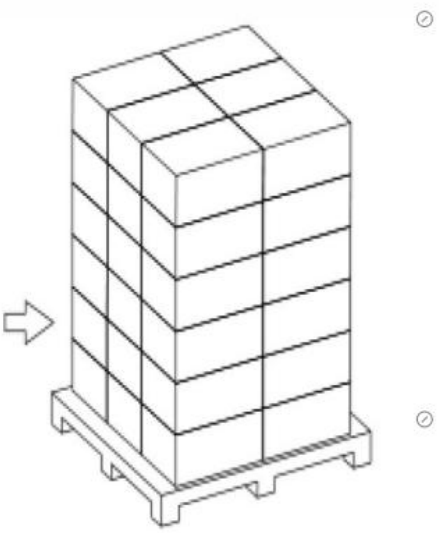
第三步：2PCS 产品放一静电袋，两片产品铁框对铁框，两袋放已卡槽每箱 120PCS



第四步，卡槽上方加珍珠棉条



第五步：封箱，贴外箱标签



第六步：每箱整齐摆放卡板上，最高堆叠 6 层

10. Precautions

Please pay attention to the followings when you use this TFT LCD module

10.1 Mounting Precautions

(1) You must mount a module using holes arranged in four corners or four sides.
(2) You should consider the mounting structure so that uneven force is not applied to the module. And the case on which a module is mounted should have sufficient strength so that external force is not transmitted directly to the module
(3) Please attach the surface transparent protective plate to the surface in order to protect the polarizer. Transparent protective plate should have sufficient strength in order to the resist external force.
(4) You should adopt heat radiation structure to satisfy the temperature specification.
(5) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the polarizer at high temperature and the latter causes circuit break by electro-chemical reaction
(6) Do not touch, push or rub the exposed polarizers with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of polarizer for bare hand or greasy cloth.
(7) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach front or rear polarizers. Do not use acetone, toluene and alcohol because they cause chemical damage to the polarizer.
(8) Wipe off saliva or water drops as soon as possible. Their long time contact with polarizer causes deformations and color fading.
(9) Do not open the case because inside circuits do not have sufficient strength
(10) When handling the LCD module, it needs to handle with care not to give mechanical stress to the PCB and Mounting Hole area."

10.2 Operating Precautions

(1) The spike noise causes the mis-operation of circuits. It should be lower than following voltage : V=± 200mV(Over and under shoot voltage)

(2) Response time depends on the temperature.(In lower temperature, it becomes longer.)

(3) Brightness depends on the temperature. (In lower temperature, it becomes lower.) And in lower temperature, response time(required time that brightness is stable after turned on) becomes longer.

(4) Be careful for condensation at sudden temperature change. Condensation makes damage to polarizer or electrical contacted parts. And after fading condensation, smear or spot will occur.

(5) When fixed patterns are displayed for a long time, remnant image is likely to occur.

(6) Module has high frequency circuits. Sufficient suppression to the electromagnetic interference shall be done by system manufacturers. Grounding and shielding methods may be important to minimized the interference.

10.3 Electrostatic Discharge Control

Since a module is composed of electronic circuits, it is not strong to electrostatic discharge. Make certain that treatment persons are connected to ground through wrist band etc. And don't touch interface pin directly..

10.4 Storage

When storing modules as spares for a long time, the following precautions are necessary.

(1) Store them in a dark place. Do not expose the module to sunlight or fluorescent light. Keep the temperature between 25±5℃ at normal humidity.

(2) The polarizer surface should not come in contact with any other object.it is recommended that they be stored in the container in which they were shipped.

10.5 Handling Precautions For Protection Film

(1) When the protection film is peeled off, static electricity is generated between the film and polarizer. This should be peeled off slowly and carefully by people who are electrically grounded and with well ion-blown equipment or in such a condition, etc.
(2) The protection film is attached to the polarizer with a small amount of glue. If some stress is applied to rub the protection film against the polarizer during the time you peel off the film, the glue is apt to remain on the polarizer. Please carefully peel off the protection film without rubbing it against the polarizer.
(3) When the module with protection film attached is stored for a long time, sometimes there remains a very small amount of glue still on the polarizer after the protection film is peeled off.
(4) You can remove the glue easily. When the glue remains on the polarizer surface or its vestige is recognized, please wipe them off with absorbent cotton waste or other soft material like chamois soaked with normal-hexane.

10.6 Safety

(1) For the crash damaged or unnecessary LCD, it is recommended to wash off liquid crystal by either of solvents such as acetone and ethanol an should be burned up later.
(2) In the case the LCD is broken, watch out whether liquid crystal leaks out or not. If your hands touch the liquid crystal, wash your hands cleanly with water an soap as soon as possible.
(3) If you accidentally swallow liquid crystal, first, wash your mouth thoroughly with water, then drink a lot of water and induce vomiting, and then, consult a physician.
(4) If the liquid crystal get in your eyes, flush your eyes with running water for at least fifteen minutes.
(5) If the liquid crystal touches your skin or clothes, remove it and wash the affected part of your skin or clothes with soap and running water.